

Ques. HOW TO DESIGN A PULSE CODE MODULATOR AND DEMODULATOR?

Sourabh

Ans. To study and design pulse code modulation (PCM), refer to Fig. 1. The analogue signal is provided to the low-pass filter, which is used to remove noise from the audio signal. Then, the audio signal is sampled to obtain a series of sampling values. Next, the signal passes through a quantiser to quantise sampling values.

The signal passes through an encoder to encode quantisation values and then converted to a digital signal. The increasing number of bits of PCM prevents the signal from distortion, but bandwidth also increases due to the increasing capacity of data.

Block diagram for pulse code demodulation is shown in Fig. 2. While transmitting a PCM signal, it is hard to avoid noise interference. Therefore before PCM signal is sent to demodulator, a comparator is utilised to convert the signal to its original level.

PCM signal is a series of pulse-wave signals. Hence, before demodulating, the series pulse-wave signal is converted to parallel digital signal by a serial-to-parallel converter.

Then, the signal passes through an n-bits decoder (normally D/A converter) to recover the digital signal to its original quantisation value. However, this value not only includes the original audio signal, it also includes a lot of high-frequency harmonics. Therefore a low-pass filter is used to remove the unwanted signal at the final stage.

For details on pulse code modulator and demodulator circuits, please refer <http://site.iugaza.edu.ps/mabuhasna/files/lab2.pdf>

Q2. WHICH LOW-COST RF WIRELESS MODULES ARE READILY AVAILABLE IN THE MARKET?

Pamarthi Kanakaraja

A2. Please find below some links for low-cost, long-range, high-quality wireless RF module products that

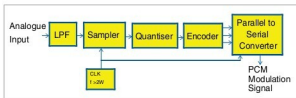


Fig. 1: Block diagram of pulse code modulation

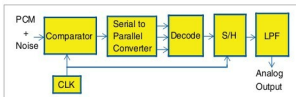


Fig. 2: Block diagram of pulse code demodulation

are most often used in medium- and low-volume products for consumer applications such as garage door openers, wireless alarm systems, industrial remote controls, smart-sensor applications and wireless home automation systems. You may select a low-cost wireless RF module as per your requirement.

www.electroniccomponents.com/RF-Transmitter-Receiver-433Mhz?filter_name=RF%20transmitter-receiver
www.raviyp.com/embedded/154-low-cost-and-long-range-rf-module-1200-meter-1-2km-in-india
www.alibaba.com/showroom/

wireless-low-cost-rf-module.html

Q3. CAN YOU EXPLAIN THE CONCEPT OF SOUND-CANCELLATION HEADPHONES AND PROVIDE LINKS FOR THEIR CIRCUIT DIAGRAMS?

Vipul

A3. There are several methods of noise cancellation in headphones. The idea behind majority of designs involves placing a microphone on the outside of the headphones to pick up ambient noise. This noise is then inverted and played into the ear with music so that you get reduced unwanted ambient sound.

Other designs contain more complex feedback systems such as those using a microphone on the inside of the headphones to determine the level of noise cancellation that is required.

Additionally, digital designs that incorporate adaptive filtering are prevalent in many commercial products today.

Noise-cancelling headphones come in either active or passive types. For detailed information, please find below a link for the circuit diagram of sound-cancellation headphones.

www.instructables.com/id/Analog-Noise-Cancelling-Headphones/electronics.howstuffworks.com/gadgets/audio-music/noise-cancelling-headphone3.htm

Answers compiled by EFY senior application engineer, Nidhi Kathuria. Letters and questions for publication may be addressed to Editor, Electronics For You, D-97/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: edfrc@efy.in) and should include name and address of the sender